Experiment No.:10

Aim:Create a simple text-based CAPTCHA and try breaking it using OCR tools (like Tesseract) to understand AI-based attacks.

In [1]:
```
#!sudo apt-get update && sudo apt-get install -y tesseract-ocr
!pip install pillow opencv-python pytesseract
```

```
Requirement already satisfied: pillow in ./anaconda3/lib/python3.9/site-packag
es (9.2.0)
Requirement already satisfied: opencv-python in ./anaconda3/lib/python3.9/site
-packages (5.0.0.93)
Requirement already satisfied: pytesseract in ./anaconda3/lib/python3.9/site-p
ackages (0.3.13)
Requirement already satisfied: numpy>=2 in ./anaconda3/lib/python3.9/site-pack
ages (from opencv-python) (2.0.2)
Requirement already satisfied: packaging>=21.3 in ./anaconda3/lib/python3.9/si
te-packages (from pytesseract) (21.3)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in ./anaconda3/lib/pyt
hon3.9/site-packages (from packaging>=21.3->pytesseract) (3.0.9)
```

In [2]:
```
!pip install --upgrade numpy pandas
```

```
Requirement already satisfied: numpy in ./anaconda3/lib/python3.9/site-package
s (2.0.2)
Requirement already satisfied: pandas in ./anaconda3/lib/python3.9/site-packag
es (2.3.3)
Requirement already satisfied: pytz>=2020.1 in ./anaconda3/lib/python3.9/site-
packages (from pandas) (2022.1)
Requirement already satisfied: tzdata>=2022.7 in ./anaconda3/lib/python3.9/sit
e-packages (from pandas) (2026.5)
Requirement already satisfied: python-dateutil>=2.8.2 in ./anaconda3/lib/pytho
n3.9/site-packages (from pandas) (2.8.2)
Requirement already satisfied: six>=1.5 in ./anaconda3/lib/python3.9/site-pack
ages (from python-dateutil>=2.8.2->pandas) (1.16.0)
```

In [3]:
```
import sys
sys.modules['pandas'] = None  # Force pytesseract to skip loading pandas
import os
import random
import cv2
import pytesseract
from PIL import Image, ImageDraw
```

In [4]:
```
width, height = 150, 50
image = Image.new("RGB", (width, height), color=(255, 255, 255))
draw = ImageDraw.Draw(image)
captcha_text = "".join(random.choices("ABCDEFGHJKLMNPQRSTUVWXYZ23456789", k=4)
draw.text((20, 15), captcha_text, fill=(0, 0, 0))
raw_filename = "captcha.png"
image.save(raw_filename)
print(f"Generated CAPTCHA text: {captcha_text}")
```

```
Generated CAPTCHA text: K3QW
```

In [5]:
```
img = cv2.imread(raw_filename, cv2.IMREAD_GRAYSCALE)
_, thresh = cv2.threshold(img, 150, 255, cv2.THRESH_BINARY)
clean_filename = "captcha_clean.png"
```

```
cv2.imwrite(clean_filename, thresh)
print(f"Preprocessed image saved as: {clean_filename}")
```

```
Preprocessed image saved as: captcha_clean.png
```

In [6]:

```
clean_image = Image.open(clean_filename)
custom_config = r"--psm 7"
extracted_text = pytesseract.image_to_string(clean_image, config=custom_config
print(f"Original Text:  {captcha_text}")
print(f"OCR Extracted:  {extracted_text}\n")

if captcha_text == extracted_text:
    print("Result: CAPTCHA successfully broken!")
else:
    print("Result: CAPTCHA resisted the attack (OCR mismatch).")
```

```
Original Text:  K3QW
OCR Extracted:  wsav

Result: CAPTCHA resisted the attack (OCR mismatch).
```

In [ ]: